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SABLE HARBOR — PROJECT CRADLE CLOSEOUT

Record ID: SH-CRD-CLOSEOUT-20260906

Version: 1.0.1

Decision date: September 6, 2026

Status: LOCKED canon except where explicitly marked IMPLEMENTATION or OPEN_NONBLOCKING

Scope: Project Cradle origin, Stream 17, first U.S. recovery deployment, Bedford, operating boundary, material genealogy, 2026 state, and supersession of CRD-008 / CRD-010 open language.

1. Authority and supersession

This record closes the remaining Project Cradle canon questions after review against current repository canon through main commit 29e8d63a9d9df17fe7d00010244385a669b70f98.

Within its stated scope this record supersedes older provisional or OPEN language, especially:

- CRD-008 customer placeholder **Gunns**;
- CRD-010 first customer/material/economics OPEN state;
- geographic working language that treated **Belle / Kanawha Valley** as the intended Cradle development site.

Earlier records remain preserved as provenance. They do not override this closeout.

The closeout does **not** convert Cradle into a separate legal subsidiary, close enterprise-wide legal-entity issue #18, close business-line export issue #12, or close the enterprise-wide Alexandria interface issue #38.

2. Cradle remains an REE-recovery business

LOCKED

Project Cradle remains the Sable Harbor business/capability for recovering rare-earth-element value from process streams that another operator already creates or must already manage.

Its business thesis remains:

Find valuable rare-earth material that existing operations already concentrate unintentionally, then add the smallest recovery intervention that captures value without breaking the host operation.

Cradle is not a conventional mine owner, not a general critical-minerals holding company, and not a vertically integrated magnet manufacturer.

The existing doctrine remains controlling:

Do not break the host system.

and:

The orebody is not the business boundary.

3. Stream 17 — first commercial reference case

LOCKED

The provisional Gunns identity is replaced by **Kelly Gang Mining**.

Kelly Gang Mining is a fictional Tasmanian industrial-minerals / heavy-mineral operator. Its existing physical separation circuit creates a low-volume side stream internally designated **Stream 17**. The stream contains a materially enriched monazite-bearing heavy-mineral fraction even though the overall host feed is not an attractive standalone rare-earth orebody.

Cradle does not redesign the host mine or require the host to adopt a new mine thesis. It adds a bounded recovery step to Stream 17.

The Cradle intervention is designed so that:

- Kelly Gang Mining retains operating authority over the host plant;
- host production does not depend on Cradle equipment being available;
- Cradle equipment can be bypassed without stopping the host process;
- Cradle owns and maintains the recovery equipment used for the reference case;
- Cradle receives an exclusive recovery right to the designated Stream 17 material during the agreement term;
- title to recovered Cradle material transfers at the defined capture point;
- Cradle bears its direct recovery operating costs;

- Kelly Gang Mining receives negotiated participation in realized recovered-product value;
- Cradle sells a **mixed REE mineral concentrate** to a specialist downstream refiner rather than separating individual rare-earth oxides or manufacturing magnets at the host site.

The first arrangement is **modestly profitable on a direct basis and immaterial at Sable Harbor enterprise scale**. Exact price, percentage participation, tonnage and margin are intentionally not fixed as permanent lore numbers; quantitative case models must derive them from explicit feed, grade, recovery, product, price and cost assumptions rather than reverse-engineering a required revenue allocation.

This resolves CRD-010 at the canon level.

4. Maeve Donnelly and the material-genealogy chain

LOCKED

Wallaby remains a killed project. Maeve does not rescue it.

Maeve's decisive contribution begins when an experienced host operator casually describes the physical process in a way that conflicts with the official flowsheet and administrative stream categories. Maeve treats the discrepancy as evidence rather than operator error.

She reconstructs the actual material genealogy across:

source → physical process → stream → sampled lot → disposition

The reconstruction shows that the existing sampling scheme follows administrative categories rather than the physical process that created concentration. The revised understanding changes Ned Kelly Mark II from a grade-finding exercise into a process-genealogy search.

The lesson survives Wallaby's death and becomes Cradle operating discipline.

For commercial recovery work, the genealogy extends through:

source → host process → designated stream → sample/assay → recovery run → captured intermediate → Bedford batch → outbound product → downstream acceptance → host settlement → financial record

Every link preserves provenance, time, uncertainty, transformation and disposition. A later analytical or financial record may summarize the chain but may not erase it.

5. First U.S. deployment — Demotte

LOCKED

Cradle's first U.S. reference deployment is an acid-mine-drainage recovery installation in north-central West Virginia operated in partnership with the fictional **Demotte Reclamation Services**.

Demotte operates ordinary treatment for water from a legacy underground coal complex. Environmental treatment and discharge obligations remain with the host operator. Cradle does not acquire the mine, the remediation liability or the host treatment system.

The first-generation Cradle unit is a containerized / skid-based **Gen 1 recovery module** connected to a controlled slipstream ahead of ordinary treatment.

Its interface rules are:

- the normal treatment path remains available at all times;
- a hard bypass returns the slipstream to ordinary treatment if Cradle equipment is unavailable or outside agreed operating limits;
- the host owns immediate water-treatment and compliance consequences;
- the host may stop or bypass Cradle when operating or environmental conditions require it;
- Cradle owns its recovery equipment, recovery procedure, maintenance and direct operating cost;
- Cradle's recovery right applies only to material captured through the defined intervention and does not create ownership of the host mine, water system or remediation asset;
- recovered intermediate is segregated, identified and transferred into Maeve's genealogy chain.

The Gen 1 deployment works but fouls more than expected under variable field chemistry. The result is not treated as failure or hidden as maintenance noise. It directly drives Gen 2 toward better continuous characterization, maintainability and operating-range awareness.

6. REE product boundary

LOCKED direction

The Appalachian AMD opportunity is economically interesting because the recovered basket can contain meaningful magnet-critical and heavy/middle-REE value, including **dysprosium, terbium and yttrium**, with **neodymium and praseodymium** recovered as part of the broader basket where present.

Cradle does not claim that every deployment has the same element distribution. Feed chemistry remains site- and time-dependent and is measured rather than assumed.

In the 2026 operating model:

- field units produce a traceable REE-rich intermediate;
- Bedford upgrades and blends recovered material into a saleable mixed rare-earth concentrate / mixed-oxide product suitable for a specialist refiner;
- Cradle does **not** commercially separate every individual rare-earth element in-house;
- Cradle does **not** manufacture permanent magnets.

Future deeper separation is an investment option, not current canon.

7. Bedford

LOCKED location and role; exact parcel IMPLEMENTATION

The Cradle U.S. recovery-development center is named **Bedford**.

Bedford is located in the **Fairmont area of north-central West Virginia** on a fictional redeveloped industrial brownfield. The working site scale is approximately **15–20 acres**.

Bedford is not a mine and is not the Demotte treatment site. It is the central development, integration and upgrading facility supporting distributed recovery deployments.

Bedford's 2026 role includes:

- analytical and process-development laboratory work;
- field-unit assembly, maintenance and refurbishment;
- controlled receipt and genealogy verification of recovered intermediates;
- blending and upgrading into saleable mixed REE products;
- secure material storage and outbound logistics;
- operator / deployment support and autopsy work on failed or degraded recovery runs.

Exact property polygon, street address, river/rail access, barge requirement and parcel history remain **implementation details for the geospatial package**, not unresolved Cradle business canon. The prior Belle / Kanawha Valley site direction is superseded.

8. 2026 operating state

LOCKED

Cradle is an **early-commercial business**, not a mature mining conglomerate and not merely a research project.

By the closeout date it has:

- the completed Wallaby failure history;
- the Stream 17 commercial reference case with Kelly Gang Mining;
- one operating U.S. Gen 1 reference deployment at Demotte;
- Bedford as its U.S. development and upgrading center;
- additional candidate streams and deployments under disciplined evaluation;
- a high kill rate by design rather than a requirement to convert every target into a project.

Kenji Arakawa remains Cradle leader. Tessa Quinn, Luis Ortega and Maeve Donnelly retain the founding roles already established in canon. This closeout does not manufacture a large department chart or new named executives merely to make the business look mature.

9. Institutional interfaces

LOCKED boundary

Cradle owns its operating and commercial decisions inside enterprise authority boundaries.

Other Sable Harbor capabilities support but do not command it:

- **Foundry** represents sources, claims, material identity, provenance, uncertainty and change;
- **Willow** may support bounded experimental work where qualification is appropriate;
- **Atlas Meridian** may investigate represented evidence but does not own Cradle deployment, contract, capital or operating decisions;
- **Alexandria / Pinakes** preserves Cradle institutional memory and makes entitled business-line material discoverable without creating a new legal entity or command channel;
- **Finance** owns authoritative financial representation, while Cradle owns the underlying operating judgment inside delegated authority.

10. Quantitative and software treatment

IMPLEMENTATION — NOT NEW PERMANENT LORE NUMBERS

The existing recovery data model and accounting flow remain useful implementation scaffolding. Their synthetic host code, allocation-driven revenue calibration and generic run values do not become Cradle canon merely because they are executable.

A Cradle quantitative successor should derive recovery economics from explicit business drivers:

- eligible feed volume;
- measured composition / grade;
- availability and operating range;
- recovery efficiency;
- recovered product specification;
- accepted quantity;
- realized price;
- host participation;
- direct operating cost;
- maintenance and consumables;
- Bedford upgrading/logistics cost;
- working-capital timing;
- capital deployment and replacement.

Historical finance releases remain reproducible. A successor must use an explicit replacement bridge rather than double-counting old allocation-based Cradle activity and new bottom-up activity.

This implementation work is covered by the broader finance / standalone-unit workstream and is **not a remaining Cradle canon question**.

11. Closeout status

CRADLE CANON CLOSED

After this record:

- CRD-001 through CRD-010 have a controlling answer;
- Gunns is superseded by Kelly Gang Mining;
- Stream 17 has a defined host, material class, product boundary and commercial structure;
- Maeve's Wallaby discovery beat and material-genealogy doctrine are defined;
- the first U.S. deployment, Gen 1 boundary and learning event are defined;
- Bedford's name, role and regional location are defined;
- Cradle's 2026 maturity is defined;
- prior Belle / Kanawha Cradle geography is superseded.

No additional owner decision is required to close Project Cradle canon.

Remaining enterprise-wide implementation or governance work must not be relabeled as an unresolved Cradle canon decision merely because Cradle is one consumer of it.

Kelly Gang Mining visual identity — September 7

The KGM-VIS-001 artwork decision locks the owner-selected Ned Kelly/two-pickaxe logo for the existing Stream 17 host operator. It changes no company relationship, host authority, recovery right or financial assumption in this closeout.

Demotte naming and artwork — September 7

DEM-RS-001 replaces the former Morrow Run company name with Demotte Reclamation Services as an editorial correction and records the owner-directed logo. This version incorporates that naming decision; all host, recovery, financial and operating boundaries remain unchanged.